Week 6: Privacy-Preservation Techniques for LLMs

Comprehensive Guide to Differential Privacy, Federated Learning, LoRA, and Embedding Defenses

AIM 5014: Special Topics in AI Safety & Security

Katz School of Science and Health
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The Privacy Challenge in Generative AI

Why Generative Models Leak Data

Large Language Models are trained to consume and memorize massive web datasets. Because they are designed to fit text patterns, they inadvertently memorize sensitive individual information.

- **The Core Risk**: Unlike traditional databases that store structured records, neural networks store information distributed across billions of weights.
- **Corporate Liability**: Enterprises fine-tuning models on medical records, legal discovery, or client emails face severe regulatory risks if data leaks.

Attack Vector 1: Training Data Extraction

Extracting Memorized Records

Adversaries craft targeted prompts that trick models into regurgitating exact training samples.

- Models frequently memorize unique strings, such as credit card numbers, phone numbers, and home addresses.
- If a prompt matches high-memorization regions of the training corpus, the model reproduces verbatim plaintext.

Attack Vector 2: Membership Inference Attacks (MIA)

Was My Record in the Training Set?

An attacker attempts to determine whether a specific individual's personal record was used to train a target language model.

- **Loss-Based Testing**: Models typically assign lower loss (higher probability) to sentences they were trained on compared to unseen text.
- By querying the model, attackers test if a target sentence has unusually low perplexity.

Attack Vector 3: Inference-Time Leakage (RAG)

RAG and Vector Databases

Retrieval-Augmented Generation systems inject private enterprise documents into model prompts at runtime.

- If vector stores or retrieval pipelines are insecure, unauthorized users can query private data repositories.
- Furthermore, dense embeddings stored in databases can be inverted back into raw text!

Differential Privacy: The Gold Standard

Mathematical Privacy Guarantee

Differential Privacy (DP) ensures that an algorithm's output does not reveal whether any single individual's data was included in the training set.

- **Neighboring Datasets**: Imagine two datasets differing by exactly one person's record.
- **Algorithmic Blindness**: A DP training process produces nearly identical model weights whether your record is present or absent.

Decoding the DP Budget: ϵ and δ

Privacy Budget (ϵ)

- Controls the strictness of the privacy bound.
- **Smaller** ϵ: Stronger privacy, but more noise.
- **Larger** ϵ: Weaker privacy, better utility.

Failure Probability (δ)

- A tiny probability that the privacy guarantee fails.
- Set strictly smaller than $1/(\text{Dataset Size})$.

Enforcing Privacy: DP-SGD (Step 1)

Per-Sample Gradient Clipping

Standard gradient descent updates parameters based on batch averages. To protect privacy, we must clip individual contributions.

- Compute gradients independently for each training example.
- Clip their L_2 norm to a maximum threshold C:

$$\bar{g}(x_i) = g(x_i) / \max\left(1, \frac{\|g(x_i)\|_2}{C}\right)$$

- Prevents any single record from dominating the gradient.

Enforcing Privacy: DP-SGD (Step 2)

Gaussian Noise Injection

After clipping individual gradients in a batch, we sum them and add calibrated random noise.

- Noise is drawn from a Gaussian distribution: $\mathcal{N}(0, \sigma^2 C^2 I)$.
- This noise masks the presence of any single individual in the training batch.
- **The Trade-off**: Higher noise protects privacy better but degrades model learning.

Du et al. (2025): DP Fine-Tuning Challenges

The High-Dimensional Noise Hurdle

Applying DP-SGD to billion-parameter LLMs introduces severe computational and utility bottlenecks.

- **Utility Collapse**: Adding noise across billions of parameters often destroys model syntax and zero-shot reasoning.
- **Hyperparameter Sensitivity**: Success requires meticulous tuning of learning rates, batch sizes, and clipping norms.

Du et al. (2026): DP Subspace Fine-Tuning

Overcoming the Noise Bottleneck

Du et al. (2026) show that LLM adaptation does not need to update every parameter.

Subspace Optimization:

- Restricts training gradients to a low-dimensional intrinsic subspace ($D_{\text{sub}} \ll D$).

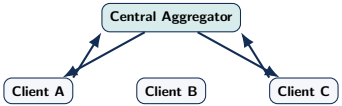
The Privacy Benefit

Lower dimensionality means exponentially less Gaussian noise is required, preserving model utility!

Federated Learning: Collaborative Training

Decentralized Intelligence (Yao et al., 2024)

Instead of collecting private user data onto a central server, Federated Learning brings the model to the data.



Clients train locally and share only lightweight weight updates ($\Delta\theta$).

Federated Learning with LLMs

Architectural Obstacles (Yao et al., 2024)

Scaling federated learning to giant language models presents major hurdles:

- **Communication Bandwidth**: Transmitting multi-gigabyte gradient updates across client connections is slow.
- **Client Heterogeneity**: Devices vary drastically in memory, compute, and data distributions (non-IID data).
- **Pre-Training Limits**: Federated pre-training remains impractical; FL is used almost exclusively for fine-tuning.

Security Risks in Federated Learning

Sharing Weights Is Not Risk-Free

Federated Learning protects raw text, but shared weight updates can still leak data.

Gradient Inversion:

- Adversarial servers can reconstruct raw training text from shared gradients.

Model Poisoning:

- Malicious clients can inject backdoors into global model weights during aggregation.

Parameter-Efficient Fine-Tuning as a Shield

Freezing the Backbone

Methods like Low-Rank Adaptation (LoRA) freeze pretrained model weights and train only small adapter matrices.

Privacy Benefits:

- Limits model capacity to memorize sensitive private tokens.
- Drastically reduces memory overhead.

DP-LoRA (Hugging Face, 2024)

Combining LoRA with differential privacy enables secure adaptation on enterprise hardware.

Mechanics of Differentially Private LoRA

Low-Rank Decomposition with Privacy

Weight updates are represented as $W = W_0 + BA$, where only A and B are trained.

- **Targeted Gradients**: Gradients are computed exclusively for adapters A and B, ignoring backbone weights.
- **Minimal Noise Variance**: Because adapter dimensions are tiny ($r \ln d$), the noise injected for DP causes minimal degradation.
- **Federated Synergy**: Clients upload only tiny LoRA adapter files rather than full model weights.

Privacy-Preserving Data Sanitization

Pre-Training Data Scrubbing

Before data enters an LLM pipeline, sanitization aims to remove or disguise sensitive attributes (Lukas et al., 2023).

Traditional Masking:

- NER tools replace names and IDs with placeholders.
- Flaw: Brittle; context often re-identifies masked entities.

Advanced Transformation (Meehan et al., 2024)

- Uses automated paraphrasing and semantic distortion.
- Preserves training utility while destroying re-identification vectors.

The Limits of Text Anonymization

Why Scrubbing Alone Fails (Lukas et al., 2023)

Adversaries exploit linguistic style, writing quirks, and secondary context to reconstruct records.

- **Stylometric Fingerprinting**: An author's unique vocabulary and syntax act as a biometric signature even when names are removed.
- **Contextual Inference**: LLM reasoning can infer missing private details from surrounding text.
- **Takeaway**: Data sanitization must be paired with algorithmic training protections like DP-LoRA.

The RAG Privacy Threat: Vector Databases

Retrieval-Augmented Generation Architecture

Production RAG systems convert private documents and medical records into dense numerical vectors stored in vector databases.



The Dangerous Myth

Developers often assume embedding vectors act as irreversible cryptographic hashes. **They do not!**

Embedding Inversion: Vec2Text (Morris et al., 2023)

Text Embeddings Reveal (Almost) As Much As Text

Morris et al. (2023) proved that dense text embeddings can be inverted to reconstruct original text with startling accuracy.

The Inversion Mechanism:

- Given encoder ϕ and vector $e = \phi(x)$, an iterative neural decoder reconstructs x.

Alarming Results

- Recovers 32-token inputs with a **97.3 BLEU score**.
- Recovers **92% of clinical medical notes exactly**!

Security Fallout of Embedding Inversion

Vector Stores Are Plaintext Leakage Vectors

Because embedding models are invertible:

- Storing unencrypted embeddings in cloud vector stores exposes sensitive data.
- Intercepting API queries or obtaining read access lets attackers extract full medical histories and source code (Morris et al., 2023).
- **Conclusion**: Embeddings require the same security controls as raw text.

Zhang et al. (2026): Securing RAG Latent Spaces

Embedding-Reversal Defenses

To protect vector databases against attacks like Vec2Text, modern architectures use latent space obfuscation.

Obfuscation Strategies:

- Add structured adversarial noise in latent space that preserves semantic search distance while breaking text decoders.

Utility Balance:

- Defenses must prevent exact token inversion without ruining cosine similarity for retrieval search (Li et al., 2024).

BinaryShield: Cross-Service Threat Intelligence

Privacy-Preserving Fingerprints (Gill et al., 2026)

Securing LLM services against coordinated attacks using cryptographic fingerprints.

- **Threat Intelligence Sharing**: Allows enterprise LLM services to share threat indicators (e.g., prompt attacks) without exposing customer data.
- **Secure Multi-Party Computation**: Enables cross-service threat verification while maintaining absolute client-side privacy.

Synthesis: The Multi-Layer Privacy Stack for LLMs

Defense-in-Depth for Generative AI

Protecting user privacy requires controls across every tier of the AI stack:

1. **Data Tier**: Advanced text transformation and sanitization.
2. **Training Tier**: Differentially Private Subspace Fine-Tuning (DP-LoRA) and Federated Learning.
3. **Storage Tier**: Latent-space obfuscation and vector database encryption for RAG.

Discussion Prompt for Class

"If embeddings can be inverted with 92% exact match accuracy, how can enterprise RAG architectures ever be deployed safely in healthcare?"

Week 6 Summary Key Takeaways

Reviewing Core Concepts

- Memorization in LLMs creates severe privacy liabilities through extraction and membership inference.
- Differential Privacy (DP-SGD, DP-Subspace) provides mathematical protection during training.
- Federated Learning enables collaborative training without sharing raw data, though it has gradient inversion risks.
- PEFT and LoRA reduce parameter footprints and minimize privacy leakage.
- Vector embeddings are invertible via Vec2Text, requiring robust latent space defenses in RAG.

Week 6 References Further Reading (Part 1)

- Du, H., et al. (2025). Can Differentially Private Fine-Tuning LLMs Protect Against Privacy Attacks? *DBSec 2025*.
- Du, L., et al. (2026). Differentially Private Subspace Fine-Tuning for Large Language Models.
- Yao, Y., et al. (2024). Federated Large Language Models: Current Progress and Future Directions.
- Morris, J. X., et al. (2023). Text Embeddings Reveal (Almost) As Much As Text: Vec2Text Inversion Attacks. *EMNLP 2023*.
- Zhang, et al. (2026). Towards Secure Retrieval-Augmented Generation: Latent Space Exploitation and Embedding Reversal Defenses.

Week 6 References Further Reading (Part 2)

- Zhang, et al. (2023). Building the Bridge Between Federated Learning and Large Language Models.
- Google Research (2024). Federated Learning Meets LLMs: Private Collaborative Fine-Tuning.
- Zhao, et al. (2023). Differentially Private Low-Rank Adaptation Using Federated Learning.
- Huang, et al. (2024). Privacy in Fine-Tuning Large Language Models: Attacks, Defenses.
- Lukas, et al. (2023) Meehan, et al. (2024): Data Sanitization and Privacy-Preserving Transformation.

Review: Why Memorization Occurs

The Overfitting Nature of Neural Networks

LLMs are massive overparameterized function approximators.

- When trained on web data containing unique sequences (like phone numbers or SSNs), the network assigns parameters specifically to predict those rare tokens.
- This exact memorization is what enables data extraction attacks.

Review: Membership Inference Mechanics

Detecting Data Presence

MIA attackers do not need the model to output raw text; they only need black-box query access.

- By measuring how confident a model is about a sentence, attackers infer if the sentence was in the training set.
- High confidence = likely member; low confidence = non-member.

Review: The Role of Epsilon in DP

Balancing Privacy and Utility

The privacy budget ϵ is a slider between safety and performance.

- Setting ϵ too low adds excessive noise, rendering the model incoherent.
- Setting ϵ too high leaves the model vulnerable to training data extraction attacks.

Review: Gradient Clipping in DP-SGD

Bounding Influence

Gradient clipping ensures no single training sample can disproportionately swing the model weights.

- By scaling down individual gradients that exceed norm C, we limit the maximum impact of any single user's data.

Review: Why Subspace Training Helps DP

Dimensionality Reduction for Noise

Adding noise to billions of parameters creates massive variance.

- Subspace tuning restricts updates to thousands of parameters instead of billions.
- Less noise is needed, resulting in much better model accuracy.

Review: Federated Learning Architecture

Decentralized Training Benefits

Data never leaves local devices or corporate servers.

- Eliminates centralized data collection liabilities.
- Complies strictly with data residency laws (e.g., GDPR, HIPAA).

Review: LoRA Adaptation Efficiency

Low-Rank Parameter Efficiency

LoRA injects small trainable matrices while freezing the base model.

- Reduces trainable parameters by over 99%.
- Minimizes memory overhead and restricts memorization capacity.

Review: The Fallacy of Text Masking

Beyond Simple Redaction

Masking names and IDs is insufficient to stop privacy leaks.

- Stylometric fingerprints and surrounding context allow re-identification.
- Robust pipelines require algorithmic privacy mechanisms.

Review: The Invertibility of Embeddings

Embeddings Are Not Cryptographic Hashes

Dense vectors retain rich semantic and syntactic structure.

- Iterative decoders can translate numerical vectors back into readable text.
- Vector databases require encryption and latent space defenses.

Next Class: Oct 14

LLM Prompt Injection (Direct & Indirect)
Required Reading/Viewing

- EchoLeak: Zero-Click Prompt Injection in M365 Copilot (Zenity Labs)
- Data Exfiltration via Indirect Prompt Injection in Bing Chat (Embrace The Red)
- Google Bard Markdown Injection & Data Exfiltration (Embrace The Red)
- Universal and Transferable Adversarial Attacks on Aligned Language Models (Zou et al., 2023)
- Prompt Injection Attacks Against Vision-Language Models in Oncology (Clusmann et al., Nature Communications 2025)
- Agent Smith: A Single Image Can Jailbreak One Million Multimodal LLM Agents (Gu et al., 2024)